

Revised Generation Interconnection System Impact Study Report

For Queue Project Q16-0705

Chattanooga Psc 100 kV
October 17, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0705. The Interconnection Customer proposes a 88.4 MW solar photovoltaic facility interconnecting at the Chattanooga Psc 100 kV substation in the Michigan area. The request was received on November 23, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 345 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$789,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

- Facility: Almena Substation - Portola 100kV
Contingency: Palisades Substation - Oceanside Substation 500kV
Loading: 106.2% (Rating: 115 MVA)
DFAX: 49.0%
Pre-project loading: 68.6%
- Facility: Sabanera 500 kV/230 kV Transformer
Contingency: Lonetree - Sabanera 500kV
Loading: 100.2% (Rating: 1,130 MVA)
DFAX: 5.3%
Pre-project loading: 99.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Almena - Portola 100kV	Reconductor line	\$978,000	48% (\$464,000)
Sabanera 500 kV/230 kV Transformer	Replace transformer with higher-rated unit	\$22,172,000	2% (\$325,000)
	Total allocated to project		\$789,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.